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Artificial intelligence and multi-omics integration in Alzheimer's disease: translational challenges

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Abstract

The past two decades have witnessed a revolution in high-throughput omics technologies, which now offer unprecedented views of the molecular complexity underlying Alzheimer's Disease (AD) and associated biomarkers. By leveraging nonlinear models and pattern recognition, artificial intelligence (AI) can integrate omics data to transcend the limitations of individual modalities, improving early detection and molecular subtype identification, expediting the identification of possible therapies via the inference of molecular pathways, and tracking disease progression. Yet, despite these breakthroughs, most AI-omics applications remain confined to research settings. From this perspective, we focus on three critical translational challenges: algorithmic bias, interpretability and trust, and the implementation of AI-driven omics into clinical practice. This perspective advocates for inclusive data infrastructures, improving interpretability and trust by providing greater mechanistic insight, and improving clinical translation by increasing reproducibility and cost-effectiveness. Such lessons from AD will seek to inform translational medicine across diverse diseases, beyond AD.

Keywords: *Alzheimer's disease, artificial intelligence, multi-omics, machine learning, deep learning, biomarkers, precision medicine*

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1. Introduction

AI has revolutionised the integration of omics datasets to allow for predictions of AD risk and the identification of biological pathways that lead to new diagnostics and therapeutics. These omics datasets include genomics, transcriptomics, proteomics, and radiomics.

Through AI, our understanding of the pathogenesis of AD can extend beyond traditionally studied genes and proteins such as Apolipoprotein (APOE) ε4, which have long been considered a strong risk factor for the development of late-onset AD. In genomics, there are examples of AI applications in genome-wide association study (GWAS) analyses, such as Bracher-Smith et al.'s [1] roadmap of utilising machine learning models to successfully unearth novel loci and complex gene interactions. Another example is the work of Ghose et al. [2], who used data from the UK Biobank and genome-wide association neural networks to identify genes and their associated biological pathways linked to family history of Alzheimer's disease. Traditional GWAS have similarly identified how variants in genes such as BIN1, CLU, and PICALM are associated with the progression from mild cognitive impairment (MCI) to AD, demonstrating how genetic data can inform disease risk and transition at early stages [3]. This would pave the way to identifying targets for drug development by linking statistically associated loci whose products influence disease biology with causal genes.

Beyond genomics, Beckmann et al. [4] demonstrated how multiscale causal networks were used to organise different scales of data across DNA, RNA, protein, and clinical data on a late-onset AD dataset, to identify gene- or protein-expression traits predicted to modulate network states driving AD. As such, the models predicted VGF as a key regulatory protein that had not been previously shown to be causally associated with AD. Further experimental validation on mouse models demonstrated that VGF was a key protector against the progression of AD. Crucially, these studies showed that the use of AI in analysing large multi-omics datasets can help to identify complex interactions and narrow down candidate genes in driving the progression of AD for downstream experimental validation.

Systematic reviews have evaluated different multi-omics approaches in AD using AI technology. Ren et al. [5] showed that combinatorial AI approaches across various omics data outperform single-omics analyses by revealing novel biomarkers through a more integrative understanding of various biological pathways. It also enables therapeutic targets to be uncovered by targeting specific molecular profiles, but these advantages are less apparent when data types are analysed in isolation. This analysis provides a better understanding of the mechanics of AD and identifies both successes and bottlenecks in moving AD-driven multi-omics tools to clinical use. Similarly, Latifi-Navid et al. [6] combined proteomics and transcriptomics data to identify early biomarkers of AD using Large Language Models (LLM) to find and access valid databases on AD and associated genes, such as KEGG, GeneCards, and the Alzheimer's Disease Variants Portal, followed by identifying network topology parameters using Cytoscape version 3.10.1 before reconstructing interaction networks [6]. This enabled identification of thirteen key genes as important

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contributors to the progression of both early and late AD, such as APP, YWHAE, YWHAH, SOD1, and UQCRFS1, as well as their effects on oxidative phosphorylation, synaptic transmission, and metabolic pathways [6].

However, despite these advances, most AI-omics applications are still far from clinical usage due to the limited predictive accuracy and the need for extensive data across diverse populations. Additionally, translating complex interaction patterns between genes into information to be used in patient care is also challenging, since AD is a polygenic disease that comprises linear and non-linear interactions between hundreds and thousands of genes, often also with small additive effects. Such challenges are especially significant in AD, a highly polygenic and heterogenous disease shaped by both linear and non-linear interactions across many genetic pathways. In this perspective, we therefore focus on three major translational challenges: algorithmic bias, interpretability and trust, and clinical translation into practice.

2. Prognostic and theragnostic challenges beyond diagnosis

In contrast to diagnosis, there are AD-specific prognostic and theragnostic AI models that can predict the rate of decline from mild cognitive impairment (MCI) to AD and response to therapies using specific biomarkers. These models provide greater insight into a patient's disease progression beyond diagnosis. Event-based models such as the probabilistic generative model used by Young et al. [7] were trained on ADNI biomarker data to explore biomarker changes as AD progresses, predicting rates of conversion from MCI to AD and offering prognostic insight. More recently, a predictive prognostic model developed by researchers at the University of Cambridge stratified patients according to slowly and rapidly progressing AD before detecting treatment effects of specific therapies such as lanabecestat on a group level—patients with slow-progressing AD showed 46% slower cognitive decline, but fast-progressing AD patients showed negligible improvement [8]. Therefore, while AI-driven prognostic models are increasingly interpretable and improve the efficiency of AD clinical trials by grouping patients according to disease progression, their translation into theragnostic tools capable of predicting an individual patient's response to specific treatments for AD may be hindered due to the need for long-term data and the vast heterogeneity of AD molecular phenotypes.

3. Algorithmic bias

Most large AD datasets come from a few well-curated cohorts, such as the Alzheimer's Disease Neuroimaging Initiative (ADNI). These data banks provide valuable information spanning omics such as genomics and transcriptomics, as well as neuroimaging and analysis of cerebrospinal fluid (CSF) biomarkers. Funded specialised centres, such as the Knight Alzheimer Disease Research Centre (ADRC) at Washington University, have also generated phenotype multi-omics biodata over 40 years of recruiting research volunteers to participate in these studies. This has enabled omics analysis in the brain, cerebrospinal fluid, and blood [9]. However, most of such genetic data have been obtained from individuals with European and North American descent, with lower representation from Asian, African, and Hispanic populations. This demographic imbalance could result in algorithmic bias and may hinder the development of personalised approaches to AD diagnostics and therapeutics in individuals of those underrepresented ethnicities [10].

This is supported by a scientometric review of genome-wide associated studies, where approximately 85% of AD GWAS participants have European ancestry, whereas only 9.92% of participants were Asian, and 1.96% were African American or Afro-Caribbean [11]. Polygenic risk scores derived from European GWAS have also been shown to perform up to fivefold less accurately in African-ancestry populations and misclassify disease risk in Asian cohorts [12]. Therefore, AI-omics models trained on European-ancestry cohorts may systematically favour those populations, giving biassed outputs, leading to potential problems in diagnosis.

Deep learning machine models trained on biassed datasets lead to misclassifications, delayed diagnoses, and the exclusion of underrepresented groups in clinical trials due to ancestry-dependent omics differences. In genetics, allele frequencies and variant effects differ substantially between European and Asian cohorts. For example, the APOE ε4 allele of the apolipoprotein E gene, a key regulator of lipid transport and cholesterol metabolism, is the strongest genetic risk factor identified for late-onset Alzheimer's disease and is also associated with increased cardiovascular risk [13]. However, the APOE ε4 allele confers a weaker risk effect for AD in African Americans compared to Europeans, a pattern further supported by analyses in admixed populations such as the Brazilian cohort analysed by Naslavsky et al. [14]. As noted by Rajabli et al. [15], other genes associated with increased African-ancestry-specific risk of AD include the gene loci 17p13.2 (which codes for Kinesin Family Member 1C (KIF1C), Profilin 1 (PFN1), and Misshapen Like Kinase 1 (MINK1)) and 18q21.33 coding for BCL2, of which the downregulation leads to an increase in autophagy in AD brains and neuronal apoptosis. Therefore, biassed data sets primarily trained on European and North American ancestry can result in diagnosis problems, such as overestimating the risk of AD in African-ancestry individuals when applying risk estimates derived from European ancestry. Clinical trials may also include heterogeneous ancestry backgrounds without stratifying for local ancestry, which can lead to contaminating effect estimates and reduced interpretability of therapies [16].

Proteomic and metabolic profiles may also be skewed based on cultural and environmental diversity across cohorts, including dietary patterns. In a HELIOS multi-ethnic Asian cohort, habitual consumption of region-relevant foods was strongly associated with distinct plasma metabolomic panels [17]. Wu et al. [18] highlighted how various metabolites showed variations in visceral adiposity and fasting glucose across ethnic groups, such as amino acid-related metabolites involved in tryptophan and histidine metabolism, and variations in seven different classes of lipid species. This means that AI-generated predictions of conversion from MCI to AD trained on European metabolomic patterns may not accurately estimate the AD risk when applied to East Asian or African populations.

Under-representation of diverse populations in omics research will lead to narrower findings, perpetuating inequity. Ancestry groups differ in genetic makeup, as well as the effect of this on the onset and progression of AD. For example, the ABCA7 gene codes for proteins involved in regulating the processing and clearance of Aβ amyloid protein, a key component of the amyloid plaques found in AD [19]. ABCA7 has a stronger association with AD risk in individuals of African ancestry than in those of European ancestry. As a result, limiting the representation of diverse populations in AD research risks missing out on new biological pathways that may be weaker in Eurocentric datasets, leading to missed opportunities to develop novel therapeutics targeting such pathways to treat populations with AD. In the case of the ABCA7 gene, an under-representation of African American cohorts may potentially lead to an underestimation of the contribution of ABCA7, obscuring its importance as a population-specific risk locus.

Meanwhile, biassed molecular models and targeted therapeutics that are less effective on Asian and African populations also raise concerns about equity and distributive justice. Individuals from underrepresented populations receive less therapeutic research, and therefore lesser therapeutics widen health disparities. Additionally, research that prioritises data from Western populations reinforces inequities in scientific visibility and healthcare access. As approval from regulatory agencies such as the FDA and EMA requires consistent performance across subpopulations, ancestry bias would pose not only ethical but also regulatory barriers to model approval, limiting the effectiveness of clinical application.

Working towards equitable AI-driven omics research into AD requires coordinated data-level and algorithmic interventions. In order to achieve more inclusive data cohorts, we can call for the building of multi-omics cohorts in Asia and Africa in order to capture the genetic, epigenetic, metabolic, and environmental heterogeneity of AD across diverse populations. For example, the Shenzhen Bay Laboratory launched the Greater Bay Area Healthy Ageing Brain Study (GHABS), representing China's efforts to restructure and design standard AD community cohorts in South China [20]. The harmonisation of data collection protocols such as the GHABS and the ADNI will allow for the creation of global biobanks which comprise data collected from diverse populations, enabling omics research to be founded on non-biassed data.

Additionally, we can work towards achieving equity in AI-omics research through the adoption of more cost-effective research techniques. For example, CSF screening is invasive and limits large-scale screening, especially in low-income countries where individuals are excluded from screening, as PET and CSF biomarker screening are too expensive and inaccessible [21]. Therefore, there is an important need to identify cost-effective biomarkers rather than the classic pathophysiological hallmarks of AD that are only detected by neuroimaging and CSF techniques. Good examples include deep learning models that leverage blood-based biomarkers such as glial fibrillary acidic protein (GFAP) that offer non-invasive and cost-effective alternatives that are suitable for deployment in less affluent societies [22]. Implementing such models will allow AI-omics to be more inclusive.

4. Interpretability and trust

Secondly, AI-driven risk scores and predictive calculations are often less explainable. Consequently, AI-based deep-learning classifiers often remain 'black boxes', as they produce predictive scores without specifying molecular or imaging features leading to their derivations. From a translational perspective, this constrains interpretability and adoption of AI-driven outputs by clinicians for therapeutic escalation, as clinicians often find it difficult to use these results to guide clinical treatment. Regulators are also unable to evaluate the biological plausibility or safety of such decisions if trialled on patients without feature attribution. As interpretability is a prerequisite for clinical translation, the validation of biological mechanisms and reproducibility across cohorts is critical.

AD exemplifies the challenges of interpretability, as clinicians are already struggling with a vast range of heterogeneous phenotypes. Hence, it is important to distinguish mechanistic drivers from correlative noise in omics-driven AI. This is necessary to avoid the generation of MCI-to-AD risk scores for a patient without clarity on how it was derived, such as from various omics signatures.

The large size of omics datasets also heightens the challenges of interpretability. To handle complex data, researchers use deep autoencoders and graph neural networks to compress data into latent variables [23]. However, such high-dimensional data with obscure biological meanings rarely provide meaningful interpretations of AD pathology. Consequently, discoveries by the AI model would be difficult to validate experimentally or develop into drug targets without explainability. From a translational standpoint, the lack of validation would be a challenge for further therapeutics, especially with lower explainability, although there would be ways to limit it. Explainable artificial intelligence (XAI) helps in demystifying the decision-making processes of AI and therefore provides potential for the translation of AI-driven omics data from bench to bedside.

Currently, approaches to explainable AI (XAI) include post hoc feature attribution methods such as SHapley Additive exPlanations (SHAP) and LIME to identify influential genes or proteins driving predictions. Shapley values are based on cooperative game theory and assign each gene or protein a score contributing towards a prediction [24]. This enables models to pinpoint specific alleles or proteins predicted as key drivers of AD. However, as noted by Vimbi et al. [24], although XAI tools such as LIME and SHAP enhance the transparency of prediction, many studies still struggle with small datasets, inconsistent explanations, and the lack of biological explanations. SHAP showed that APOE4 was often not a top driver in The Alzheimer's Disease Prediction Of Longitudinal Evolution (TADPOLE) models, but was only relevant when imaging information was removed, despite its clinical relevance to AD. This highlights how models may prioritise downstream phenotypes while ignoring upstream molecular drivers of pathology, which are more essential in disease progression.

Another approach is attention mechanisms, which also weigh different inputs, highlighting which omics features or brain regions the AI model is concentrating on when making its inference [25]. However, attention maps are constrained because attention weights reflect correlation but may not accurately reflect causality.

A key point of diagnostic relevance is the combination of XAI techniques such as saliency maps and grad-weighted class activation mapping (grad-CAM), providing insights targeted to specific regions of the brain known to exert an influence on AD pathology [26]. These approaches are attractive in neuroimaging because they allow for voxel-level visualisation of model attention, potentially aligning model predictions with established substrates of disease. Nevertheless, these saliency maps can be unstable and noisy, as small changes in input or model parameters can lead to drastic changes in saliency maps [27]. Such sensitivity raises concerns regarding the robustness and reproducibility of explanation methods and affects reliability in radiomics and neuroimaging. For example, Eitel and Ritter conducted a systematic robustness analysis comparing multiple attribution methods. In the analysis, identical convolutional neural network architectures were trained multiple times using different random initialisations, and attribution maps were generated using the different attribution methods. These methods included gradient × input, guided backpropagation, layer-wise relevance propagation (LRP), and perturbation-based occlusion approaches. The robustness of each attribution method was quantified by measuring the similarity between relevance maps obtained across different training runs using spatial and quantitative consistency metrics. They noted that gradient-based attribution methods performed the worst, as they yielded relevance maps with the lowest consistency across different runs [28]. Hence, causal inference and counterfactual explanations distinguishing mechanistic factors from correlates are arguably the greatest weakness of existing AI models.

Such approaches can hence be used to identify which properties are responsible for model predictions linking computational accuracy and biological plausibility, particularly when the AI output can be used to formulate hypotheses for further experimental testing by researchers in the field. Once researchers verify biological plausibility in the laboratory from these approaches, clinical translation is then possible. While such tools offer hope in being more closely linked to causation rather than correlation, they are still potentially constrained by instability.

For trust to develop, clinicians must trust outputs by AI models, regulators must trust reproducibility, and patients must trust that their data are used responsibly, as well as predictions made by the AI model. In AD, this is especially sensitive, as a black-box classifier labelling a patient "high risk" for dementia carries profound psychological and social implications, which must be explained carefully and adequately to the patient. The lack of transparent reasoning with the patient may cause harm, eroding trust between clinician and patient rather than providing meaningful help.

This has been recognised by regulatory agencies. The SaMD AI/ML framework published by the FDA highlights the importance of enhancing the transparency and explainability when developing AI models [29]. Clinician oversight is another prerequisite for approval. The European Medicines Agency (EMA)'s reflection paper on AI stresses the need for a balanced collaborative approach that prioritises the safety of patients and preserves regulatory integrity. For omics-driven AD models, this means that predictions must be accurate and grounded in pathophysiological features that are interpretable by clinicians, such as APOE status or neuroinflammatory markers in the progression of MCI to AD.

We argue that explainability in AI-driven omics needs to develop into actionable interpretability, which enables clinicians to translate explanations by the AI model to patients. While current methods provide some level of transparency, as discussed above, they remain unstable and potentially disparate from our understanding of causal biology. Future directions include embedding interpretability into model design. Biology-constrained models, which build biological knowledge directly into their design rather than learning purely from data, can align latent features with molecular pathways such as complement activation or lipid metabolism, ensuring mechanistic plausibility. Moreover, clinician-facing dashboards that display weighted contributions by each omic dataset would make outputs more comprehensible and translatable in practice. Finally, explanations must remain consistent across datasets and ancestries. Vlontzou et al. [30] illustrate various interpretability methods, including "feature attribution techniques and counterfactual explanations" to generate "human-friendly interpretations on the outputs" of the best-performing AI models for MCI and AD diagnosis, providing reason for optimism that clinician adoption would depend on ongoing efforts to ensure actionable interpretability with biological validation.

5. Clinical translation and adoption

Currently, promising signature AD-associated biomarkers have been identified. GFAP, a type III intermediate filament protein specific to astrocytes, has been characterised as a biomarker for injury to astrocytes [31]. Neurofilament Light Chain (NfL), a marker of axonal degeneration [32], and tau proteoforms such as pTau-217, which arise from pathological post-translational hyperphosphorylation [33], can be used to track the progression of AD. These biomarkers are detectable by CSF and plasma assays. When integrated with AI models, they show early promise in diagnosis. However, the use of AI models for predictions of these biomarkers remains largely confined to the research stage, with none approved for AD by either the FDA or EMA, highlighting translational barriers.

Concerns surrounding the reproducibility in external datasets are a key limiting factor for regulatory approval, which requires prospective and reproducible evidence, as argued earlier. Variability arises from differences in pre-analytical handling, assay platforms, population demographics, and comorbidities. Regulatory approval is therefore challenging without robust multi-site validation and standardised reference ranges. For instance, while NfL levels are consistently elevated across dementias, their overlap with other neurological diseases undermines specificity for AD.

Moreover, high-throughput multi-omics integration methods with the use of AI-omics assays are often less cost-effective than already-established imaging or CSF biomarkers, limiting the scalability of AI-driven omics data in diagnosis [34]. For example, obtaining high-throughput omics data is far more expensive per subject through methods such as whole-genome sequencing, quantitative proteomics, and targeted metabolomics compared to routine biomarkers such as MRI, PET, or CSF Aβ/tau. Moreover, with the lack of harmonised sequencing or proteomics platforms, this may pose another challenge, as regulatory agencies require the use of standardised protocols for approval.

There are several generalisable lessons from the translational hurdles facing AI-omics in AD, which we can draw from other fields. In oncology, multi-gene assays such as Oncotype DX analyse the activity of 21 genes that can influence how likely a cancer is to grow and respond to treatment, guiding breast cancer treatment by predicting the responsiveness to chemotherapy to treat early-stage breast cancers [35]. It categorises the expression of genes into favourable genes, such as the oestrogen receptor group, which algorithmically predicts a lower likelihood of recurrence and greater response to chemotherapy. In contrast, unfavourable genes such as human epidermal growth factor (HER)-2 group, invasion group, and CD68 correspond to a higher recurrence scale in the algorithm. However, these tests required decades of trials and validation before achieving FDA clearance. Today, many candidate genomic signatures remain unapproved, reflecting the high evidentiary bar.

In cardiology, similarly, current artificial intelligence-based omics models have generated relatively accurate predictions for cardiovascular disease. For example, Degroat et al. [36] showed that 18 transcriptomic biomarkers were highly significant in the CVD population and hence were used to predict disease with up to 96% accuracy when cross-validated with the clinical records of patients. Different machine learning algorithms aggregate in this approach to stratify the risk for heart disease.

These similarities and differences point towards three overarching recommendations. Firstly, multi-centre prospective validation as part of the approval process is necessary. Solely relying on retrospective cross-validation cannot suffice, as prospective validation needs to be undertaken in real-time across various hospitals to validate that these AI tools work satisfactorily despite the range of presenting symptoms and underlying mechanisms. An interesting approach would be the utilisation of electronic medical record (EMR)-driven multi-omics data, as outlined by Wu et al. [37], which allows for the constant refinement and recalibration of AI models, improving the validity of outputs. Secondly, cost analyses must prove additional value to existing standards to gain approval as a new diagnostic standard that is cost-effective for populations. Thirdly, methods such as standardised pipelines are imperative to promote trust and regulatory approval.

However, the challenge faced by AI-omics translation stems not from a lack of discovery but from the challenge of validation that already exists within oncology and cardiology, as previously discussed. Most of these would likely hold true for the neurodegeneration community. The challenge of closing that gap begins with aligning model development with the same regulatory standards as clinical trials.

We will highlight some key developments that will accelerate clinical translation and adoption. Firstly, prospective validation trials should be done to embed AI-omics models in longitudinal AD studies. Secondly, cost-effectiveness analyses should be done to compare multi-omics profiling by AI models with current diagnostics for AD (e.g., cognitive testing, MRI, PET imaging, CSF Aβ/tau assays) in terms of health outcomes such as early detection of AD and prediction of disease progression. Thirdly, to increase generalisability, representation from multi-ethnic cohorts should be maintained, and various databases such as the UK Biobank, EPAD, and ADNI should be standardised. Fourthly, to increase interpretability, AI model outputs should be biologically sound to be trusted by clinicians for clinical application to patients. Finally, integration with clinical workflow and electronic health records would increase accessibility and utilisation by clinicians, improving translation and adoption.

6. Advantages and practical considerations of AI-driven multi-omics analyses

The use of AI models in AD offers many advantages. Firstly, in genomics, machine-learning approaches applied to GWAS can use stable input data to identify novel risk loci and non-linear relationships between gene variants that may not be captured by conventional single-variant statistical analyses. This allows the identification of potentially novel targets for therapeutics and highlights key gene-gene interactions that reflect the polygenic nature of AD, where many variants contribute small additive effects to the risk of disease. Secondly, in transcriptomic and proteomic analyses, AI models are effective at unravelling the biological mechanisms that drive various subtypes of AD. By detecting hidden causal relationships within datasets, AI models can uncover various AD molecular subtypes and predict biological pathways driving AD heterogeneity, providing insights that could not have emerged from the sole use of traditional differential expression analyses.

Despite these benefits, the practical application of AI in AD faces multiple challenges. Beyond the accuracy of AI models, the application of these tools would depend on duration, feasibility, data requirements, and interpretability. Across various modalities, translational AI models would require long-term follow-up with patients to assess their accuracy and reliability. This could potentially increase the duration and cost of studies. Moreover, machine learning models still require large ancestry-diverse cohorts, which could limit clinical deployment. Additionally, metabolic models remain sensitive to environmental and lifestyle factors, which could complicate the standardisation and reliability of the AI models. Comparatively, imaging-based AI models and deep learning models utilising 3D MRI data potentially offer the greatest translational potential, as neuroimaging of AD is clinically accessible, but this remains costly and data-intensive.

7. Conclusions

In conclusion, the convergence of artificial intelligence and the use of multi-omics data offers unprecedented potential to unravel the complexities driving AD progression, but translation remains limited due to biassed data, opaque algorithms, and fragmented validation, as discussed. To ensure equitability and regulatory approval for clinical translation, we call for diverse populations to be embedded in data infrastructures which are grounded in biologically plausible mechanisms and must be reproducible. This will pave the path from discovery to deployment in clinical practice for AD and wider neurodegenerative disorders.

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Author contributions

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Conflict of interest

The authors declare that they have no competing interests.

Data availability statement

All data supporting the findings of this publication are available within this article.

Additional information

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